[METRIC]
MIL-PRF-62382B(AT)
19 October 1995
SUPERSEDING
MIL-V-62382A(AT)
26 March 1986

PERFORMANCE SPECIFICATION

VALVE, HYDRAULIC, PRESSURE RELIEF (METRIC)

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of in-line hydraulic pressure relief valve used in a military combat vehicle to control system differential pressure at a predetermined level (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for addition information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4820

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-H-5606	- Hydraulic Fluid, Petroleum Base: Aircraft, Missile and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base: For Preservation and Operation.

STANDARDS

FEDERAL

FED-STD-595	- Colors Used in Government Procurement.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

DEPARTMENT OF DEFENSE

12265558	- Valve, Pressure Relief.

(Copies of this drawing are available from the U.S. Army Tank-automotive and Armaments Command, Warren, MI 48397-5000.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z1.4 - Sampling Procedures and Tables for Inspections by Attributes.

(Copies of ASQC Standards may be obtained from The American Society for Quality Control, P.O. Box 305, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117 - Standard Test Method of Salt Spray (Fog) Testing.

(Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for hydraulic valves. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable.

3.2.1 Age-sensitive materials. The use of age-sensitive materials shall be as specified in the contract or order. The maximum age of the valve poppet seal (see 3.2.5) shall not exceed 40 quarters at time of acceptance of the valve by the Government.

3.2.2 Fungus. Only materials which are fungus non-nutrient shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

3.2.3 Hydraulic fluids. The valve shall be compatible with hydraulic fluids conforming to MIL-H-5606 and MIL-H-6081.

3.2.4 Metals. All metals used in the construction of the valves except metal which is in constant contact with the hydraulic fluid, shall be of a corrosion-resistant type or shall be suitably protected to resist corrosion during the normal service life of the valves. The use of dissimilar metals, especially brass, copper, or steel in intimate metal to metal contact, shall be avoided.

3.2.5 Poppet seal. The valve poppet seal shall be of elastomer material to minimize leakage under shock and vibration conditions.

3.3 Design and construction. The valve shall conform to the manufacturer's design and construction for hydraulic pressure relief valves except for outline dimensions, and porting hole dimensions and locations which shall conform to drawing 12265558 (see 4.5.2).

3.3.1 End cap. The threaded end cap (outlet port) of the valve shall be safety-lockwired with 10.4 millimeter (mm) (0.41 inch) diameter corrosion resistant steel wire in accordance with drawing 12265558.

3.4 Performance.

3.4.1 Proof pressure. The valve shall withstand an internal pressure of 31 000 kilopascals (kPa) (4500 pounds per square inch gage [psig]) without leaking externally. The valve shall not be damaged and shall subsequently meet the requirements of 3.4.3 and 3.4.4 (see 4.5.4.1).

3.4.2 Burst pressure. The valve shall withstand an internal pressure of 52 000 kPa (7500 psig) without rupture of internal or external parts. The valves need not be operable after this exposure (see 4.5.4.2).

3.4.3 Function. Unless otherwise specified herein, the valve shall function as specified below with hydraulic pressure applied to the inlet port.

3.4.3.1 Cracking pressure. The valve poppet shall crack open when the differential pressure across the valve is not more than 140 kPa (20 psi). The flow after cracking shall be not less than 1.9 liter per minute (L/m) (0.5 gallons per minute [gpm]) at 140 kPa (20 $\pm$ 1 pounds per square inch differential [psid]) (see 4.5.4.3).

3.4.3.2 Rated flow and pressure drop. With fluid flow through the valve of 20 L/m (5 gpm), the pressure drop through the valve shall not be greater than 100 kPa (15 psid) (see 4.5.4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

3.4.3.3 Reseat pressure. After flow is established through the valve, the poppet shall reseat when the differential pressure is not less than 70 kPa (10 psid). No flow shall be permitted in the reverse direction (see 4.5.4.5).

3.4.4 Low pressure leakage. There shall be no evidence of leakage through the valve when a pressure of 35 kPa (5 psig) is applied at the inlet port; there shall be no evident leakage when 200 kPa (30 psig) is applied at the outlet port (see 4.5.4.6).

3.4.5 Environmental. The valve shall meet the requirements of 3.4.1 and shall operate as specified in 3.4.3 and 3.4.4 under the following environmental conditions.

3.4.5.1 High temperature. The valve shall operate after exposure to an ambient temperature of +70 degrees Celsius (° C) (+160 degrees Fahrenheit [°F]) and during and after exposure to +60°C (+140°F) (see 4.5.5.1).

3.4.5.2 Low temperature. The valve shall operate after exposure to an ambient temperature of -54°C (-65°F), and during and after exposure to -32°C (-25°F) (see 4.5.5.2).

3.4.5.3 Fluid temperature. The valve shall operate when exposed to hydraulic fluid temperatures ranging from +70°C (+160°F) to -32°C (-25°F) (see 4.5.5.3).

3.4.5.4 Salt fog. The valve shall operate after exposure to a salt-fog environment (see 4.5.5.4).

3.4.5.5 Shock resistance. The valve shall operate after exposure to a shock impulse of 25 gravity units (g) for 11 milliseconds (ms) +0 -1 ms, half-sine wave, applied along three mutually perpendicular axes (see 4.5.5.5).

3.4.5.6 Vibration resistance. The valve shall operate after exposure to the following vibration conditions (see 4.5.5.6).

a. Simple harmonic motion.
b. Motion applied along three mutually perpendicular axes.
c. Peak acceleration of 10g at all frequencies.
d. Frequency range, 20 to 2000 Hertz (Hz).
e. Frequency sweep varied logarithmically.
f. Sweep time, 35 $\pm$ 5 minutes.
g. Dwell 5 minutes at each observed resonant frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

3.5 Cleanliness. Upon completion of testing, sufficient quantity of fluid shall be retained in the valve to assure proper preservation during extended storage. Ports shall be covered with suitable plugs. Valve internal cleanliness shall conform to the level shown in table I (see 4.6).

Maximum Size of Surface Particles (Micrometers)	Maximum Quantity of Particles
25	7450
50	1020
100	95
250	2

3.6 Painting. The exterior surfaces of the valve shall be prepared, primed or treated, and painted as suitable for the material of the valve body (see 4.5.2). Color of finish shall be light green, matching color chip 24533 of FED-STD-595.

3.7 Marking. The valve shall be marked with the National Stock Number (NSN) and the manufacturer's name and part number. Marking shall be permanent and legible (see 4.5.2).

3.8 Workmanship. Workmanship shall be of the quality as to assure that valves furnished under this specification are free of defects that compromise, limit, or reduce performance in intended use (see 4.5.2).

4 VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6 (see table III).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

TABLE II. Classification of defects.

Category	Defect	Method of examination
Major		
101	Dimensions affecting interchangeability, not within tolerance (see 3.3).	SIE 1/
102	End cap safety-lockwiring, not to requirements (see 3.3.1).	Visual
103	Identification marking, improper (see 3.7).	Visual
104	Faulty workmanship affecting performance (see 3.8).	SIE 1/
Minor		
201	Dimensions not affecting interchangeability, not within tolerance (see 3.3).	SIE and Visual
202	Painting, improper (see 3.6).	Visual
203	Faulty workmanship affecting appearance (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance tests
Proof pressure	3.4.1	4.5.4.1	X	X
Burst pressure	3.4.2	4.5.4.2	X	
Cracking pressure	3.4.3.1	4.5.4.3	X	X
Rated flow and pressure drop	3.4.3.2	4.5.4.4	X	X
Reseat pressure	3.4.3.3	4.5.4.5	X	X
Low pressure leakage	3.4.4	4.5.4.6	X	X
High temperature	3.4.5.1	4.5.5.1	X	
Low temperature	3.4.5.2	4.5.5.2	X	
Fluid temperatures	3.4.5.3	4.5.5.3	X	
Salt fog	3.4.5.4	4.5.5.4	X	
Shock resistance	3.4.5.5	4.5.5.5	X	
Vibration resistance	3.4.5.6	4.5.5.6	X	
Cleanliness	3.5	4.6	X	X

4.2.1 Preproduction inspection. When specified (see 6.2), first article preproduction

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

inspection shall be performed on two valves. This inspection shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6.9 (see table III).

4.2.2 Initial production inspection. Unless otherwise specified (see 6.2), first article preproduction inspection shall be performed on two valves. This examination shall include the examinations of 4.4 (see table II) and the tests of 4.5.4.1 through 4.6.9 (see table III).

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.5.4.1 through 4.5.4.6 and 4.6 (see table III).

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ANSI/ASQC Z1.4. Any redesign or modification of the contractor's standard to comply with specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements of requirements of 3.3, 3.3.1, 3.6, 3.7 and 3.8 as listed in table II. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.1, 3.6, 3.7 and 3.8 shall be determined by examination for the defects listed in table II. Examination shall be visual, tactile, or by measurement with SIE.

4.5.3 Test conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 23 $\pm$ 10°C (73 $\pm$ 18°F), barometric pressure shall be 725 +50 -75 mm mercury (Hg) (28.5 +2 -3 inches Hg), and relative humidity shall be 50 $\pm$ 30 percent. Unless otherwise specified herein, the valve shall meet all specified performance requirements in a hydraulic system, utilizing hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083 or the commercial equivalents under the following conditions:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

a. Hydraulic fluid temperature of 38 ± 5°C (100 ± 10°F).
b. Hydraulic fluid operating pressure of 20 700 ± 700 kPa (3000 ± 100 psi).

4.5.4 Performance.

4.5.4.1 Proof pressure. To determine conformance to 3.4.1, the valve shall be subjected to a static pressure test of 31 000 kPa (4500 psi) for a duration of 2 minutes. There shall be no external leakage. The test shall be performed with pressure applied to the inlet port with the outlet port plugged.

4.5.4.2 Burst pressure. To determine conformance to 3.4.2, the valve shall be subjected to a static pressure test of 52 000 kPa (7500 psig) for a duration of 2 minutes. Pressure shall be applied to the inlet port, with the outlet port plugged. External or internal parts of the valve shall not rupture. This is a destructive test, so test valves shall be indelibly marked “DO NOT USE” and discarded.

4.5.4.3 Cracking pressure. To determine conformance to 3.4.3.1, pressure shall be applied slowly to the valve inlet until the valve poppet cracks open. The pressure differential at the instant of valve opening shall not be more than 140 kPa (20 psid). After opening, the flow shall be not less than 1.9 L/m (0.5 gpm) at 140 ± 7 kPa (20 ± 1 psid).

4.5.4.4 Rated flow and pressure drop. To determine conformance to 3.4.3.2, pressure shall be applied to the valve inlet and the flow shall be adjusted to 20 L/m (5 gpm). The pressure drop through the valve shall not be more than 100 kPa (15 psi).

4.5.4.5 Reseat pressure. To determine conformance to 3.4.3.3, a pressure of 172 kPa (25 psi) shall be applied to the valve inlet and flow through the valve verified. Then the pressure shall be gradually decreased until the valve closes. The differential pressure at the instant of valve closing shall be not less than 70 kPa (10 psi).

4.5.4.6 Low pressure leakage. To determine conformance to 3.4.4, alternately apply a pressure of 35 kPa (5 psi) to the valve inlet, then 200 kPa (30 psig) to the outlet port. There shall be no evidence leakage through the valve.

4.5.5 Environmental.

4.5.5.1 High temperature. To determine conformance to 3.4.5.1, the valve shall be heated in a chamber until its temperature stabilizes at +70°C (+160°F). Remove the valve from the chamber and allow it to cool to ambient room temperature. The valve shall meet the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

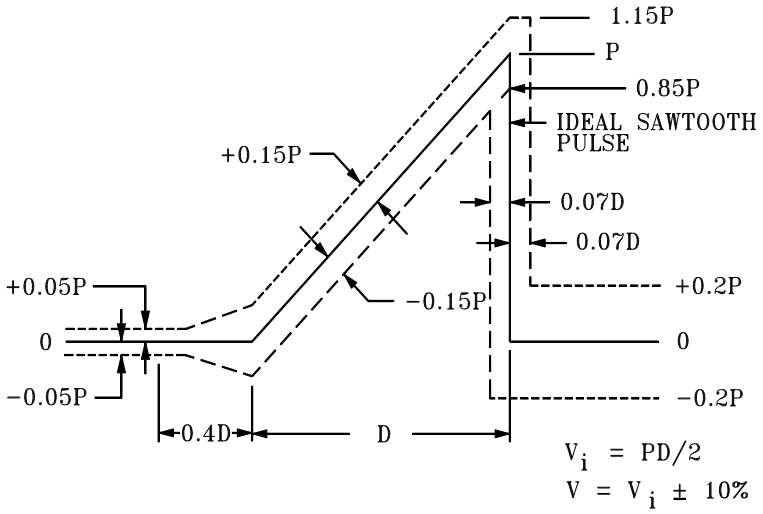
applicable requirement of 3.4.5.1. Next, heat the valve in a chamber until its temperature stabilizes at +60°C (+140°F). The valve shall meet the applicable requirement of 3.4.5.1 at the stabilized temperature and after it is cooled to ambient room temperature.

4.5.5.2 Low temperature. To determine conformance to 3.4.5.2, the valve shall be cooled in a chamber until its temperature stabilizes at -54°C (-65°F). Remove the valve from the chamber and allow it to warm to ambient room temperature. The valve shall meet the applicable requirement of 3.4.5.2. Next, cool the valve in a chamber until its temperature stabilizes at -32°C (-25°F). The valve shall meet the applicable requirement of 3.4.5.2 at the stabilized temperature and after it is warmed to ambient room temperature.

4.5.5.3 Fluid temperatures. Conformance to 3.4.5.3 shall be determined during testing in accordance with 4.5.5.1 and 4.5.5.2.

4.5.5.4 Salt fog. To determine conformance to 3.4.5.4, the valve shall be exposed to a salt fog environment in accordance with ASTM B117.

4.5.5.5 Shock resistance. To determine conformance to 3.4.5.5, the valve shall be subjected to a half sine shock pulse as shown in figure 1.



NOTE:
The oscillogram should include a time about 3D long with the pulse located approximately in the center. The integration to determine velocity change should extend from .4D before the pulse to .1D beyond the pulse. The acceleration amplitude of the ideal half sine pulse is A and its duration is D. Any measured acceleration pulse which can be contained between the broken line boundaries is a nominal half sine pulse of nominal amplitude A and nominal duration D. The velocity-change associated with the measured acceleration pulse is V.

FIGURE 1. Tolerances for half sine shock pulse.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

4.5.5.6 Vibration resistance. To determine conformance to 3.4.5.6, the valve shall be subjected to a high frequency vibration as shown in figure 2.

4.6 Cleanliness. To determine conformance to 3.5, the following shall be performed:

a. Prior to any performance testing the internal cleanliness of the valve shall be established by providing operating pressure at the level specified in 4.5.1 and then establishing a flow between the two ports. Maintain the flow until the effluent meets the cleanliness of table I.

b. At the conclusion of quality conformance testing, re-establish the assemblies internal cleanliness by performing the same flushing and effluent check. Once the required cleanliness has been achieved, drain the effluent and seal the assembly with a non-contaminating, tamper proof seal, adequately tagged and identified, before submitting it to preparation for delivery or storage processing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

VIBRATION FREQUENCY (HZ)

$G = .0512f^2DA$ (f - frequency in hertz, DA = double amplitude in mm)

FIGURE 2. Vibration - test curve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve, furnished under this specification, is a component part of the hydraulic control system of the M901 Improved Tow Vehicle (ITV) launcher and the M981 Fire Support Team Vehicle (FISTV) targeting head.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article samples are not required (see 3.1).
d. If preproduction inspection is required (see 4.2.1).
e. If initial production inspection is not required (see 4.2.2).
f. Test conditions if other than as specified (see 4.5.3).
g. Packaging requirements.

6.3 Subject term (key word) listing.

Hydraulic fluid
Hydraulic valve
Pressure relief
Valve

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.

Custodian:
Army - AT

Preparing activity:
Army - AT

Review activity:
DLA - CS

(Project 4820-0678)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:39Z
Check the source to verify that this is the current version before use.